

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	15 July 2026
Team ID	
Project Name	UCab Premium – Cab Booking Application
Maximum Marks	4 Marks

Technical Architecture

UCab Premium is built on the MERN stack (MongoDB, Express.js, React, Node.js), organized in an MVC pattern on the backend and a component-based structure on the frontend.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	Rider, Driver, and Admin web dashboards	React (Vite), HTML, CSS, JavaScript
2.	Application Logic-1	REST API layer handling routing and requests	Node.js, Express.js
3.	Application Logic-2	Authentication and role-based access control	JSON Web Tokens (JWT), bcrypt
4.	Application Logic-3	Business logic for booking lifecycle & driver assignment	Express controllers (MVC)
5.	Database	Primary data store for users, drivers, bookings, payments, reviews	MongoDB (5 collections)
6.	Local Database Server	MongoDB instance running on localhost for development	MongoDB Community Server
7.	File/Document Generation	Trip receipt generation as downloadable PDF	PDFKit (Node.js)
8.	Analytics & Visualization	Admin dashboard charts for rides, revenue, and growth	Recharts (React charting library)
9.	External API-1	Sending confirmation and notification emails	Nodemailer / SMTP email service
10.	Machine Learning Model	Not applicable in current scope	—
11.	Infrastructure (Server/Cloud)	Application deployment target	Local Server (current); MongoDB Atlas / cloud hosting (future option)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Core frameworks used across the stack	React, Express.js, Node.js, MongoDB
2.	Security	Authentication, authorization, and data protection	JWT, RBAC, bcrypt password hashing, input validation
3.	Scalable Architecture	Modular MVC backend and separated frontend allow independent scaling	Node.js/Express MVC, REST API (35 endpoints)
4.	Availability	Backend designed for continuous operation with predictable error handling	Express error-handling middleware

5.	Performance	Efficient queries and lightweight JSON payloads over REST	MongoDB indexing, Express routing
----	-------------	---	-----------------------------------